MUTUAL INDUCTANCE OF TWO LOOPS   file: mloop.mcd

Formulas included in this spreadsheet:

Mutual inductance of two loops   MLOOP()

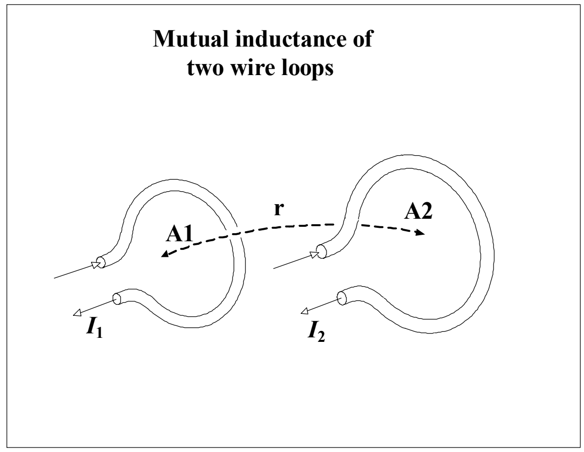
Variables used:

r   Separation between loop centers (in.)

A1 Surface area of loop 1 (in.^{2})

A2 Surface area of loop 2 (in.^{2})

(We assume the loops are flat, and that their faces are oriented parallel to each other for maximum coupling)



mloop

The loops must be well separated for the MLOOP() approximation to work:

$r > \sqrt{A1}$ and $r > \sqrt{A2}$

Mutual inductance of two well-separated loops (nH):

$$MLOOP(r, A1, A2) := 5.08 \cdot \frac{A1 \cdot A2}{r^3}$$